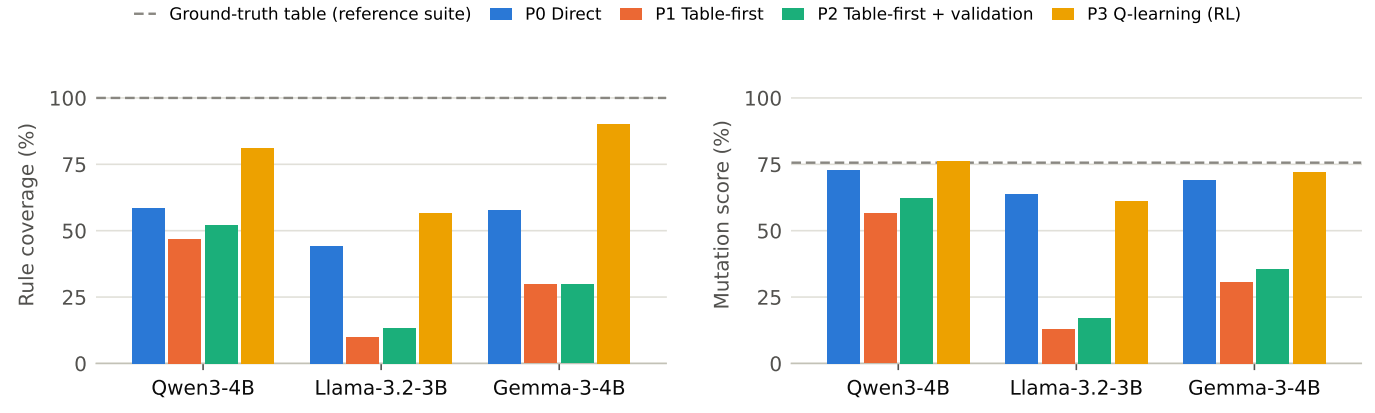
Ground-truth table (reference suite)
P0 Direct
P1 Table-first
P2 Table-first + validation
P3 Q-learning (RL)
Rule coverage (%)
100
75
50
25
0
Qwen3-4B
Llama-3.2-3B
Gemma-3-4B
Mutation score (%)
100
75
50
25
0
Qwen3-4B
Llama-3.2-3B
Gemma-3-4B